Razorpay Trusted Collections Agent

A Hindi voice agent for bounced EMIs that recommends who takes the borrower next: the AI or a person.

Ujjwal Goel · Build Challenge Track 2: Recover and Grow with AI · 13 October 2026

THE PROBLEM

For the head of collections at an NBFC: roughly one EMI auto-debit in three fails, and recovery is bought in headcount. One caller covers about 400 borrowers, so chasing more bounces means hiring more people. The decision that moves money, which borrower deserves which kind of effort, is made by feel.

India's personal loan market	Figure	Source
Loans outstanding	₹17.0 lakh crore across 12.7 crore live loans (Jun 2026)	CRIF High Mark
EMI auto-debits that fail	~29% of NACH debits by volume, 22% by value	NPCI data (public series ends 2022)
Still unpaid 31–180 days later	2.3% of the book ≈ ₹39,000 crore; 5.7% for small-ticket NBFC loans	CRIF Jun-26; rupee figure is my arithmetic

WHAT WE HEARD, AND HOW IT IS HANDLED TODAY

Three interviews at an Indian lender that runs collections in-house: a product lead, a collections analyst and a data-science lead.

- **A fixed pattern.** Bounces drop into the CRM and each agent places 900–1,000 calls a day, repeating until a positive disposition. A promise just schedules a callback. Good agents prioritise in Excel and follow up on WhatsApp, which is encouraged but not tracked.
- **Intent is judged by gut.** About 30% of non-payers have an intent problem and 70% a genuine one (medical, salary delay, family), but there is no fixed rule and it varies by agent and region. The record is a one-word disposition.
- **Quality is sampled.** A quality team hears 10–20% of calls. An LLM now scores all of them, and its dispositions match the agents' about 80% of the time.
- **The bot already works; the decision does not.** AI calling cut cost from ₹11 to ₹3–4 a minute and matched humans in an A/B test. What is still unbuilt: choosing which borrower goes to a person, and using the last call to shape the next.

THE SOLUTION

- **Calls within minutes, in Hindi and Hinglish.** The Razorpay Trusted Collections Agent reaches the borrower while the bounce is hours old.
- **Negotiates like a good agent, inside the lender's rules.** Identity first, asks why and listens, proposes a date that fits payday, one payment link, a ticket instead of a promise it cannot make.
- **Turns the call into signals, then recommends and plans the next step.** Reason, tone and promise specificity plus the borrower's history give one route: keep the AI, or hand to a person. The demo borrower makes the point: a firm promise, yet four bounces in six months send him to a person.

KEY PRODUCT AND AI DECISIONS

- **The product ends in a plan, not a recommendation.** A route for one borrower does not change collections. The Plan tab turns the routes into resource allocation for the cycle: how many AI calls, person calls and reminders, how much lands by day 1, 3 and 7, and where the gap to target sits (the agent, our callers, or where effort goes).
- **Two questions decide the route, and rules decide, not the model.** Has intent shifted (four or more bounces in the window)? Did the AI let the borrower down on this call? Either yes sends the borrower to a person. The model only extracts facts (reason, promise, tone).
- **Hard rules and judgement rules, so every verdict can be explained.** Hard rules (one link, no threat, nothing unapproved) are checked in code; judgement rules (listened first) are scored by a model against a quote from the call. AI suggested leaving all judgement to the model; I kept the split because a lender cannot audit a verdict it cannot explain.

DELIBERATELY LEFT OUT

- **Pre-bounce awareness calling.** That is about making the borrower aware in time, which an IVR call or a reminder already does. The talking happens after the bounce, and that is where capturing intent matters, so that is where we started.
- **One collections playbook for every lender.** Each bank or NBFC sets up its own agent, so the demo journey is a sample, not a standard. We provide the layer beneath any agent, better signals and the next step rather than another collections bot, so teams can build their own agent on it.

WHY RAZORPAY

- **It already runs the mandate that bounced.** A failed debit already fires a webhook, which is the trigger this agent needs, and the payment link and the measurement land on the same rails.
- **It sells an outcome, not plumbing.** Rupees recovered, priced against ₹11-a-minute human calling, under a regulated aggregator's recorded, rule-scored calls.

SUCCESS METRIC AND WHAT TO TEST NEXT

Metric	What it tells the lender	Demo build (sample of 100 EMIs)
PRIMARY: payment collected	Rupees recovered of rupees due	46% · ₹2,30,000 of ₹5,00,000
Resolution rate and time	Bounced EMIs cleared in full, and how fast	40 of 100 · 5 days median
Hard-rule breach rate	Compliance; the target is zero	29.7% — the gap to close first
Routing accuracy	Handovers a reviewer agrees with	Not yet measured; needs live calls

Next test: split one month's bounced EMIs, half through the agent and its routing and half through today's process, and compare rupees collected in nine days and whether routed-to-person cases recover more than if left with the AI. Sample figures are synthetic; weights are uncalibrated and the recommendation is shown, not executed.